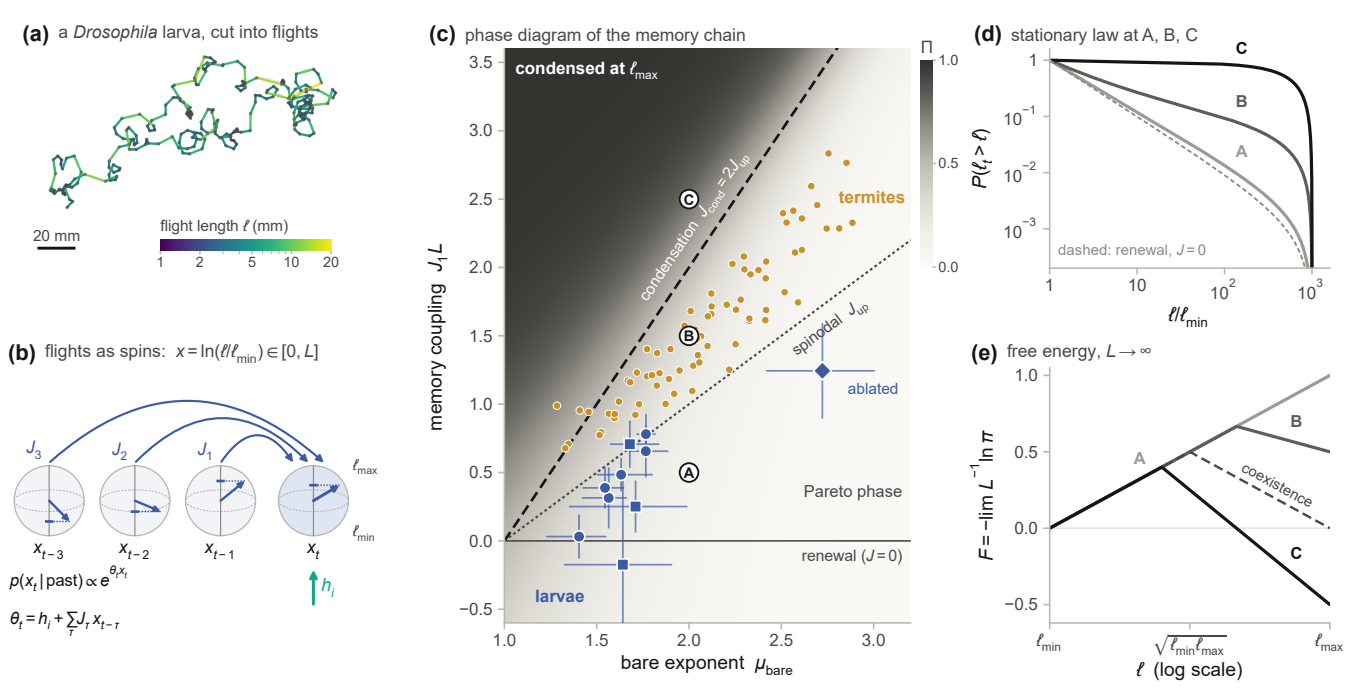

(a) a Drosophila larva, cut into flights
20 mm
flight length ℓ (mm)
1 2 5 10 20
(b) flights as spins: x = ln(ℓ/ℓmin) ∈ [0, L]
J3 J2 J1
ℓmax
ℓmin
xt−3 xt−2 xt−1 xt
p(xt | past) ∝ e^{θt xt}
θt = hi + Σ_τ Jτ xt−τ
hi
(c) phase diagram of the memory chain
condensed at ℓmax
Π
1.0
0.5
0.0
condensation J_conv = 2J_up
termites
spinodal J_up
ablated
Pareto phase
renewal (J = 0)
larvae
memory coupling J1L
bare exponent μbare
(d) stationary law at A, B, C
C
B
A
dashed: renewal, J = 0
P(ℓt > ℓ)
ℓ/ℓmin
(e) free energy, L → ∞
B
A
coexistence
C
F = −lim L^{−1} ln π
ℓmin
√(ℓmin ℓmax)
ℓmax
ℓ (log scale)